

FINAL SHANK3 ASO Mechanism-of-Action Report

R / targets chemistry-aware bioinformatics workflow

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1 Executive summary

This report summarizes a reproducible **R-only** workflow for two SHANK3-associated ASOs (ASO_4_5_2_4, ASO_4_5_2_6) in *Homo sapiens* **GRCh38**. Both ASOs are linked to **increased SHANK3 expression**. Chemistry is **phosphorothioate + 2'-MOE with all cytosines as 5-methyl-dC; architecture is unknown** and **RNase-H compatibility is unknown**. Sequence/database evidence is used only to generate **ranked hypotheses**, not confirmed mechanisms.

metric	value
Exact SHANK3 mapping rows	13
Approximate mapping rows	1
Top mechanism labels	ASO_4_5_2_4=RBP occlusion; AS...

2 ASO sequences and chemistry

aso	sequence	len	GC	backbone	MOE	m5dC	arch
ASO_4_5_2_4	TCCTGACTGTGCGCTGGG	18	38.89	phosphorothioate	T	T	unknown
ASO_4_5_2_6	AGGGCACTAAGCAATGTA	18	27.78	phosphorothioate	T	T	unknown

Full table (all columns): results/01_sequence_qc.tsv

Chemistry rules applied throughout: do not assume gapmer architecture; do not assume RNase-H activity; annotate every C as 5m-dC while retaining canonical bases for alignment/motifs.

3 Target gene and phenotype context

- Gene: **SHANK3** (neuronal synaptic scaffolding; dosage-sensitive ASD/Phelan-McDermid relevant gene).
- Cell types of interest: **neuron, neural precursor cell**.
- Phenotype for both ASOs: **increased SHANK3 expression**.
- Implication: mechanisms that **destroy** SHANK3 mRNA via RNase-H are directionally inconsistent with the stated phenotype for simple on-target knockdown (unless isoform switching/feedback is demonstrated).

4 Transcript and genomic mapping (GRCh38)

aso	match	exact	tx	chrom	pos	str	mm
ASO_4_5_2_4	exact_transcript_rc	T	ENST000004147...	chr22	5599-5616	+	0
ASO_4_5_2_4	exact_transcript_rc	T	ENST000006739...	chr22	6295-6312	+	0
ASO_4_5_2_4	exact_transcript_rc	T	ENST000002627...	chr22	5465-5482	+	0
ASO_4_5_2_4	exact_transcript_rc	T	ENST000004452...	chr22	4067-4084	+	0
ASO_4_5_2_4	exact_transcript_rc	T	ENST000006644...	chr22	3961-3978	+	0
ASO_4_5_2_4	exact_transcript_rc	T	ENST000007103...	chr22_KQ7...	5982-5999	+	0
ASO_4_5_2_4	approx_transcript...	F	ENST000006593...	chr22	919-936	+	2
ASO_4_5_2_6	exact_transcript_rc	T	ENST000004147...	chr22	5563-5580	+	0
ASO_4_5_2_6	exact_transcript_rc	T	ENST000006739...	chr22	6259-6276	+	0
ASO_4_5_2_6	exact_transcript_rc	T	ENST000002627...	chr22	5429-5446	+	0
ASO_4_5_2_6	exact_transcript_rc	T	ENST000004452...	chr22	4031-4048	+	0
ASO_4_5_2_6	exact_transcript_rc	T	ENST000006644...	chr22	3925-3942	+	0
ASO_4_5_2_6	exact_transcript_rc	T	ENST000006593...	chr22	883-900	+	0
ASO_4_5_2_6	exact_transcript_rc	T	ENST000007103...	chr22_KQ7...	5946-5963	+	0

Full table (all columns): results/02_shank3_mapping.tsv

Windows (+/- 200 / 200 nt) are written to results/02_target_windows.fasta and .bed.

5 RNA structure and accessibility

ViennaRNA (RNAfold/RNAlfold) is invoked only via `system2()`. Missing tools produce warnings and empty numeric fields - **results are never fabricated**. All energies are canonical-RNA approximations unsuitable as exact MOE/PS/5m-dC biochemical measurements.

aso	ctx	MFE	site	unpaired	hyb	status
ASO_4_5_2_4	aso_self	-0.2	NA	NA	NA	ok
ASO_4_5_2_6	aso_self	-0.6	NA	NA	NA	ok
ASO_4_5_2_4	target_window	-200	bulge_or_junc...	0.52	-38.4	ok
ASO_4_5_2_4	target_window	-200	bulge_or_junc...	0.52	-38.4	ok
ASO_4_5_2_4	target_window	-200	bulge_or_junc...	0.52	-38.4	ok
ASO_4_5_2_4	target_window	-200	bulge_or_junc...	0.52	-38.4	ok
ASO_4_5_2_4	target_window	-200	bulge_or_junc...	0.52	-38.4	ok
ASO_4_5_2_4	target_window	-200	bulge_or_junc...	0.52	-38.4	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok
ASO_4_5_2_6	target_window	-88.2	bulge_or_junc...	0.56	-30.8	ok
ASO_4_5_2_6	target_window	-199.1	bulge_or_junc...	0.52	-30.8	ok

Full table (all columns): results/03_rna_structure.tsv

6 RBP CLIP, motif, expression, and function evidence

Sources referenced: ENCODE eCLIP, UCSC ENCODE tracks, POSTAR3, oRNAment, ATtRACT, RBPDB, RBP2GO (see provenance). **No CLIP peaks are invented.**

6.1 CLIP overlaps

aso	rbp	source	chrom	site	ov0	strength	status
ASO_4_5_2_4	-	ENCODE_eCLIP;POSTAR3;...	chr22	5599-5616	F	none	metadata_only_no_peak_overlap
ASO_4_5_2_4	-	ENCODE_eCLIP;POSTAR3;...	chr22	6295-6312	F	none	metadata_only_no_peak_overlap
ASO_4_5_2_4	-	ENCODE_eCLIP;POSTAR3;...	chr22	5465-5482	F	none	metadata_only_no_peak_overlap
ASO_4_5_2_6	-	ENCODE_eCLIP;POSTAR3;...	chr22	5563-5580	F	none	metadata_only_no_peak_overlap
ASO_4_5_2_6	-	ENCODE_eCLIP;POSTAR3;...	chr22	6259-6276	F	none	metadata_only_no_peak_overlap
ASO_4_5_2_6	-	ENCODE_eCLIP;POSTAR3;...	chr22	5429-5446	F	none	metadata_only_no_peak_overlap

Full table (all columns): results/04_rbp_clip_overlap.tsv

6.2 Motif overlaps (sample)

aso	rbp	motif	start	end
ASO_4_5_2_4	FMR1	FMR1_ACYK	184	187
ASO_4_5_2_4	FMR1	FMR1_ACYK	280	283
ASO_4_5_2_4	FMR1	FMR1_ACYK	322	325
ASO_4_5_2_4	RBFOX1	RBFOX_GCAUG	123	127
ASO_4_5_2_4	RBFOX1	RBFOX_GCAUG	332	336
ASO_4_5_2_4	RBFOX3	RBFOX_GCAUG	123	127
ASO_4_5_2_4	RBFOX3	RBFOX_GCAUG	332	336
ASO_4_5_2_4	FMR1	FMR1_ACYK	184	187
ASO_4_5_2_4	FMR1	FMR1_ACYK	280	283
ASO_4_5_2_4	FMR1	FMR1_ACYK	322	325
ASO_4_5_2_4	RBFOX1	RBFOX_GCAUG	123	127
ASO_4_5_2_4	RBFOX1	RBFOX_GCAUG	332	336

Table 1: Table (continued)

aso	rbp	motif	start	end
ASO_4_5_2_4	RBFOX3	RBFOX_GCAUG	123	127
ASO_4_5_2_4	RBFOX3	RBFOX_GCAUG	332	336
ASO_4_5_2_4	FMR1	FMR1_ACYK	184	187
ASO_4_5_2_4	FMR1	FMR1_ACYK	280	283
ASO_4_5_2_4	FMR1	FMR1_ACYK	322	325
ASO_4_5_2_4	RBFOX1	RBFOX_GCAUG	123	127

Full table (all columns): results/04_rbp_motif_overlap.tsv

6.3 Expression priors (neuron / NPC)

aso	rbp	cell	expressed	status
ASO_4_5_2_4	FMR1	neuron	T	prior_annotation
ASO_4_5_2_4	ELAVL2	neuron	T	prior_annotation
ASO_4_5_2_4	ELAVL4	neuron	T	prior_annotation
ASO_4_5_2_4	RBFOX1	neuron	T	prior_annotation
ASO_4_5_2_4	RBFOX3	neuron	T	prior_annotation
ASO_4_5_2_4	PTBP2	neuron	T	prior_annotation
ASO_4_5_2_4	CELF4	neuron	T	prior_annotation
ASO_4_5_2_4	ATXN2	neuron	T	prior_annotation
ASO_4_5_2_4	TDP43	neuron	T	prior_annotation
ASO_4_5_2_4	HNRNPA2B1	neuron	T	prior_annotation
ASO_4_5_2_4	LIN28A	neuron	T	prior_annotation
ASO_4_5_2_4	PUM2	neuron	T	prior_annotation

Full table (all columns): results/04_rbp_expression.tsv

Table 2: Table (continued)

aso	rbp	cell	expressed	status
ASO_4_5_2_4	MOV10	neuron	T	prior_annotation
ASO_4_5_2_4	UPF1	neuron	T	prior_annotation

aso	rbp	process	direction	status
ASO_4_5_2_4	FMR1	translation	context_dependent	curated_annotation
ASO_4_5_2_4	ELAVL2	stabilization	activating_stab...	curated_annotation
ASO_4_5_2_4	ELAVL4	stabilization	activating_stab...	curated_annotation
ASO_4_5_2_4	RBFOX1	splicing	context_dependent	curated_annotation
ASO_4_5_2_4	RBFOX3	splicing	context_dependent	curated_annotation
ASO_4_5_2_4	PTBP2	splicing	repressive_spli...	curated_annotation
ASO_4_5_2_4	CELF4	splicing	context_dependent	curated_annotation
ASO_4_5_2_4	ATXN2	translation	context_dependent	curated_annotation
ASO_4_5_2_4	TDP43	stability	context_dependent	curated_annotation
ASO_4_5_2_4	HNRNPA2B1	processing	context_dependent	curated_annotation
ASO_4_5_2_4	LIN28A	translation	context_dependent	curated_annotation
ASO_4_5_2_4	PUM2	repression	repressive	curated_annotation

Table 3: Table (continued)

aso	rbp	process	direction	status
ASO_4_5_2_4	MOV10	remodeling	structure_related	curated_annotation
ASO_4_5_2_4	UPF1	surveillance	repressive	curated_annotation

6.4 Function annotation

Full table (all columns): results/04_rbp_function.tsv

7 Ranked RBP hypotheses

Transparent scoring: **+3 / +2 / +1 / -2** rules in R/06_mechanism_ranking.R. Top 3-5 RBPs per ASO attempt to include repressive, activating/stabilizing, and structure-related classes.

aso	rank	rbp	score	class
ASO_4_5_2_4	1	PTBP2	4	repressive
ASO_4_5_2_6	1	PTBP2	4	repressive
ASO_4_5_2_4	2	RBFOX1	4	splicing
ASO_4_5_2_6	2	RBFOX1	4	splicing
ASO_4_5_2_4	3	RBFOX3	4	splicing
ASO_4_5_2_6	3	RBFOX3	4	splicing
ASO_4_5_2_4	4	SRSF1	4	activating_st...
ASO_4_5_2_6	4	SRSF1	4	activating_st...
ASO_4_5_2_4	5	AGO2	3	repressive
ASO_4_5_2_6	5	AGO2	3	repressive

Full table (all columns): results/06_ranked_rbps.tsv

8 TF and chromatin (conditional)

TF motifs are evaluated on **genomic DNA only** when locus class suggests nuclear regulatory / intronic / promoter / enhancer / antisense context. This evidence stream is separate from RBP analysis.

aso	tf	status	warning
ASO_4_5_2_4	-	skipped_not_regulatory_context	No TF inference on RNA; step ...
ASO_4_5_2_6	-	skipped_not_regulatory_context	No TF inference on RNA; step ...

Full table (all columns): results/05_tf_chromatin.tsv

9 Chemistry-aware mechanism ranking

Eight chemistry/mechanism rules are applied (architecture unknown; RNase-H unknown; phenotype direction; PS protein engagement; MOE occupancy bias; 5m-dC caveats; structure hypothetical; sequence != proof). Labels scored as hypotheses only.

aso	rank	mechanism	score	RNaseH	top_rbps
ASO_4_5_2_4	1	RBP occlusion	8	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_4	2	RNA stabilization	3	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_4	3	splicing or processing	3	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_4	4	unresolved	3	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_4	5	RBP displacement	2	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_6	1	RBP occlusion	8	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_6	2	RNA stabilization	3	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_6	3	splicing or processing	3	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_6	4	unresolved	3	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...
ASO_4_5_2_6	5	RBP displacement	2	unknown	PTBP2,RBFOX1,RBFOX3,SRSF1...

Full table (all columns): results/07_mechanism_hypotheses.tsv

10 Per-ASO interpretation

10.1 ASO_4_5_2_4 (TCCTGACTGTGCGCTGGG)

- Top mechanism hypothesis: **RBP occlusion**
- Top RBPs: PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2
- RNase-H: **unknown** (unsupported by chemistry assumption; architecture unknown; phenotype argues against simple on-target RNase-H knockdown)
- Exact SHANK3 match present: TRUE

10.2 ASO_4_5_2_6 (AGGGCACTAAGCAATGTA)

- Top mechanism hypothesis: **RBP occlusion**
- Top RBPs: PTBP2, RBFOX1, RBFOX3, SRSF1, AGO2
- RNase-H: **unknown** (unsupported by chemistry assumption; architecture unknown; phenotype argues against simple on-target RNase-H knockdown)
- Exact SHANK3 match present: TRUE

11 Off-target considerations

A limited **chr22 genomic exact reverse-complement** screen is recorded in results/02_offtargets_chr22.tsv. This is **not** a full transcriptome off-target proof. Off-target mechanisms remain possible, especially without exact SHANK3 transcript confirmation.

12 Validation recommendations

For every high-priority RBP, the following ten experiment types are recommended (each states which hypothesis it tests):

aso	rbp	experiment	tests
ASO_4_5_2_4	AGO2	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_4	AGO2	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_4	PTBP2	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_4	PTBP2	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_4	RBFOX1	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_4	RBFOX1	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_4	RBFOX3	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_4	RBFOX3	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_4	SRSF1	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_4	SRSF1	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_6	AGO2	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_6	AGO2	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_6	PTBP2	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_6	PTBP2	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_6	RBFOX1	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_6	RBFOX1	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...

Table 4: Table (continued)

aso	rbp	experiment	tests
ASO_4_5_2_6	RBFOX3	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_6	RBFOX3	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...
ASO_4_5_2_6	SRSF1	ASO + RBP CLIP / eCLIP (or RI...	Tests whether RBP occupies th...
ASO_4_5_2_6	SRSF1	RBP knockdown / CRISPRi and m...	Tests whether RBP is necessar...

Full table (all columns): results/07_validation_recommendations.tsv

13 Resource provenance

resource	version	build	date
GENCODE_GTF	GENCODE Release 45	GRCh38	2026-09-15
TxDb.Hsapiens.U...	3.18.0	GRCh38	2026-09-15
BSgenome.Hsapie...	1.4.5	GRCh38	2026-09-15
ENCODE_eCLIP_se...	ENCODE Portal API	GRCh38	2026-09-15
POSTAR3	POSTAR3	GRCh38	2026-09-15
oRNAmnt	oRNAmnt	GRCh38	2026-09-15
ATtRACT	ATtRACT	GRCh38	2026-09-15
RBPDB	RBPDB	GRCh38	2026-09-15
RBP2GO	RBP2GO	GRCh38	2026-09-15
ENCODE_TF	ENCODE Portal	GRCh38	2026-09-15
ChIP-Atlas	ChIP-Atlas	GRCh38	2026-09-15
SCREEN_cCREs	SCREEN / ENCODE cCREs	GRCh38	2026-09-15

Full table (all columns): data/raw/PROVENANCE.tsv

14 Limitations and missing information

- ASO **architecture** not provided (cannot confirm gapmer / mixmer / fully modified).
- **RNase-H compatibility unknown**; not assumed.
- Neuron/NPC **eCLIP peak overlaps** at the precise site generally unavailable in this automated run (metadata may be cached; peaks not fabricated).
- Motif/expression layers include **explicitly labeled priors**.
- Structure values omit MOE/PS/5m-dC parameterization.
- No experimental ASO engagement assay integrated.
- Genome builds other than GRCh38 were **not** used.
- PDF tables show **readable column subsets** (layout fix); full-width result tables remain in results/*.tsv. Analysis values are unchanged.

15 Conclusions (non-confirmatory)

Both ASOs should be interpreted under chemistry-aware occupancy / RBP / structure / splicing hypotheses rather than assumed RNase-H gapmer knockdown. The highest-scoring mechanism labels are **hypotheses only**. Definitive mechanism assignment requires the validation suite above, architecture disclosure, and locus-resolved CLIP/structure/functional genetics in neuron/NPC systems.

16 Reproducibility

```
cd shank3-aso-mechanism
Rscript -e 'targets::tar_make()'
```

Pipeline orchestrated with the `targets` package; modules in R/01_*.R-R/07_*.R; config in _config.yml.